

Aritro Pathak

Phone: (+91) 91473-25939

Email: ap7mx@missouri.edu

VISITING POSITIONS

Spring 2026 Visiting Scientist, Stat-Math Unit, Indian Statistical Institute, Kolkata, India.

Fall 2025 Visiting Researcher, Tata Institute of Fundamental Research,
Center for Applicable Mathematics, Bangalore, India.

EDUCATION

Fall 2021 - Summer 2025 University of Missouri, Missouri, USA: PhD in Mathematics;

Fall 2016 - Spring 2018 Brandeis University, Massachusetts, USA: M.A. in Mathematics.

Fall 2014 – Spring 2016 Duke University, North Carolina, USA: PhD program in Physics;
36 credit hours of advanced courses in Physics and Mathematics (no degree)

Fall 2009 – Spring 2014 Indian Institute of Technology Kanpur, Uttar Pradesh, India: Undergraduate
degree in Physics

RESEARCH INTERESTS

- Geometric measure theory.
- Harmonic analysis on Euclidean spaces.
- Elliptic partial differential equations.
- Additive Combinatorics.

PUBLICATIONS

(All preprints and papers found at [Researchgate link.](#))

1. Aritro Pathak, "Marstrand type slicing statements in $\mathbb{Z}^2 \subset \mathbb{R}^2$ are false for the counting dimension", Online Journal of Analytic Combinatorics 21(2026) 1-12.
2. Aritro Pathak, "Open intervals in sums and products of Cantor sets", arxiv.org/abs/2307.08135 (Accepted in Rocky Mountain Journal of Mathematics).
3. Aritro Pathak, "Ergodicity in the linked twist maps with counter-oriented twists: ", arxiv.org/abs/2307.08124, (Accepted in Indian Journal of Pure and Applied Mathematics) <https://doi.org/10.1007/s13226-025-00844-z>.
4. Aritro Pathak. "A discrete Marstrand type slicing theorem with the mass dimension." The Electronic Journal of Combinatorics (2024): P3-13.
5. Aritro Pathak. "On certain partition bijections related to Euler's partition problem." Discrete Mathematics 345, no. 2 (2022): 112673.
6. Aritro Pathak, "Glaisher's Partition Problem" , The Mathematics Student 89 (Nos. 1-2, January-June 2020).
7. Aritro Pathak, "Reply to Comment on 'An elementary argument for the magnetic field outside a solenoid'", European Journal of Physics 40 (2), 028004 (2019).
8. Aritro Pathak, "Reals in the unit interval as average of two reals in the Cantor's middle third set", The Mathematics Student 87 (Nos 3-4 July-December 2018).
9. Aritro Pathak, "An elementary argument for the magnetic field outside a solenoid", European Journal of Physics 38 (1), 015201 (2017).
10. Aritro Pathak, "On the electrostatic equilibrium of charges and cavities in a conductor", European Journal of Physics 37 (6), 065203 (2016).

PREPRINTS

1. Aritro Pathak, "On the dimension drop for harmonic measure on uniformly non-flat Ahlfors-David regular boundaries", arxiv.org/abs/2601.16167 (Submitted to Analysis & PDE).
2. Aritro Pathak, "A counterexample for pointwise upper bounds on Green's function with a singular drift at boundary", arxiv.org/abs/2405.13313 (Submitted to Acta Mathematica Hungarica).
3. Aritro Pathak, "BMO solvability with singular drifts on ample sawtooth domains implies L^p solvability", arxiv.org/abs/2511.12747 (Submitted to Mathematika).
4. Aritro Pathak, "Doubling of elliptic measure with drifts satisfying an average diverging condition with a small constant", arxiv.org/abs/2511.12942 (Submitted to Nonlinear Analysis).
5. Aritro Pathak, "Pointwise bounds on Dirichlet Green's functions for a singular drift term", arxiv.org/abs/2511.12741 (Submitted to Communications in Partial Differential Equations).
6. Aritro Pathak, "A class of one-periodic tilings in the integer grid." (Under revision with Journal of the Indian Mathematical Society).
7. Aritro Pathak, "Ergodicity of a surgered flow on unit tangent bundle of hyperbolic surface", arxiv.org/abs/2307.08126 (Submitted to Nagoya Mathematical Journal).
8. Aritro Pathak, "Conditional upper bounds on the least character non-residue" arxiv.org/abs/2405.13316, (Under revision in Indian Journal of Pure and Applied Mathematics).
9. Aritro Pathak, "Difference sets in Quadratic Density Hales Jewett conjecture with 2 letters", [arXiv:2008.08556](https://arxiv.org/abs/2008.08556) (Submitted to Czechoslovak Mathematical Journal).
10. Aritro Pathak, "Sums of powers, and products of elements of the middle third Cantor set", [arXiv:2004.05145](https://arxiv.org/abs/2004.05145) (Under revision with The Mathematics Student).

TEACHING EXPERIENCE**University of Missouri:**

- Course Instructor, MATH 1300: Finite Mathematics (Spring 2022, Spring 2023, Fall 2023, Fall 2024, Spring 2025).
- Course Instructor, MATH 1100: College Algebra (Fall 2021, Fall 2022).
- Course Instructor, MATH 1400: Calculus for Social and Life Sciences (Spring 2024).

Brandeis University:

- Course Instructor, MATH 10A: Techniques of Calculus (a) (Fall 2017, Spring 2018, Summer 2018, Fall 2018).
- Course Instructor, MATH 10B: Techniques of Calculus (b) (Fall 2019).
- Teaching Assistant, MATH 15A: Applied Linear Algebra (Spring 2019).
- Teaching Assistant, MATH 22B: Honors Linear Algebra and Multi-variable Calculus, Part II (Spring 2017).
- Teaching Assistant, MATH 36A: Probability (Fall 2016).

Duke University:

- Teaching Assistant, PHY 362: Electricity and Magnetism (Fall 2014).
- Teaching Assistant, PHY 305: Introduction to Astrophysics (Spring 2015).
- Teaching Assistant, PHY 522: Special and General Relativity (Spring 2015).
- Teaching Assistant, PHY 464: Quantum Mechanics (Fall 2015).
- Teaching Assistant, PHY 762: Electrodynamics (Graduate course) (Spring 2016).

TALKS AND PRESENTATIONS

- June. 2025* "Green's function estimates and elliptic measures for some linear elliptic equations with singular drifts." Tata Institute of Fundamental Research-Center for Applicable Mathematics, Bangalore.
- April. 2025* "Green's function estimates and elliptic measures for some linear elliptic equations with singular drifts." PhD defense talk at University of Missouri.
- Feb. 2025* "Dirichlet solvability for elliptic equations, with a singular drift that satisfies a Carleson measure condition". Analysis Seminar, Washington University in St Louis.
- Jan. 2025* Short presentation of current and past research, Fourier Analysis and Convex Geometry group in University of Lille.
- Jan.2020* "Sums of powers of elements of the Cantor set". Graduate Student Seminar, Brandeis University.
- Nov. 2019* "Invariant measures and the set of exceptions to Littlewood's conjecture" after Einsiedler, Katok, Lindenstrauss, at Tufts Dynamics and Probability Seminar.
- Nov. 2019* "A Marstrand type slicing theorem with mass dimension in the integers", at Brandeis Combinatorics Seminar.
- Oct. 2018* "An Overview of the Green-Tao theorem". Graduate Student Seminar, Brandeis University.

SCHOOLS AND WORKSHOPS

- March 2026* (Upcoming) Cantor Set & Brownian Motion, IISER Berhampur, India.
- May 2025* Recent Advances in Harmonic Analysis and Partial Differential Equations, Baylor University.
- April 2025* Expanding Pathways in Harmonic Analysis, Washington University of St Louis.
- March 2025* Hot Topics: Interactions between Harmonic Analysis, Homogeneous Dynamics, and Number Theory, Simons Laufer Mathematical Sciences Institute (MSRI).
- May 2022* Houston Summer School on Dynamical Systems at University of Houston.
- March 2020* Analytic and Discrete Aspects of Finite Point Configurations at CIRM Luminy (Cancelled due to COVID 19).
- January 2020* Winter School on Dynamics: Topology and Numbers at HIM, Bonn, Germany
- May 2018* Houston Summer School on Dynamical Systems at University of Houston.

SERVICE TO PROFESSION

Invited as a reviewer for Mathematical Reviews, on MathSciNet.

Refereed manuscripts for *European Journal of Physics*, *Physica Scripta*, *Classical and Quantum Gravity*, all published by IOP Publishing (<https://publons.com/researcher/1475337/aritro-pathak/>) .

'IOP Trusted reviewer status' certificate (top 15% reviewers) received in Fall 2021.

MISCELLANEOUS

- Ranked 1st in the written exam for entrance to the Physics PhD program in 2014 at Tata Institute of Fundamental Research (TIFR), the leading research institute for basic sciences in India.
- Selected for the final Orientation cum Selection camp for the team to represent India in the International Chemistry Olympiad in 2009.
- The paper "An elementary argument for the magnetic field outside a solenoid", is on a Physics Stackexchange [post](#) with over 115,000 views.
- Amateur violinist in Hindustani classical and Western classical music.